

Math Review

ECON 5001
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What is Game Theory?

Game theory is the study of strategic interaction: settings where each player's best choice depends on what the others do.

- Players, strategies, and payoffs
- Solution concepts: dominance, Nash equilibrium, and refinements
- Applications across economics, political science, biology, and CS

A First Example: The Prisoner's Dilemma

Two players simultaneously choose to *Cooperate* or *Defect*:

		Player 2	
		C	D
Player 1	C	-1, -1	-3, <u>0</u>
	D	<u>0</u> , -3	<u>-2</u> , <u>-2</u>

Defect strictly dominates *Cooperate* for both players, so the unique Nash equilibrium is (D, D) — even though both prefer (C, C).

Roadmap

1. Strategic-form games and dominance
2. Nash equilibrium
3. Mixed strategies
4. Extensive-form games and backward induction
5. Repeated games and reputation